# MOSTAFA SUMAR NASER

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## 01 // EDUCATION

**Technical University of Darmstadt (TU Darmstadt)** Oct 2026 – Present

*Bachelor of Science (B.Sc.) in Computer Science — Darmstadt, Germany*

- Core Focus: Algorithms & Data Structures, Discrete Mathematics, Systems Engineering
- Coursework: Functional & Object-Oriented Programming (Java), Automata & Formal Languages

**Martin-Behaim-Schule** Aug 2025 – Jun 2026

*Fachhochschulreife (Business Information Systems) | Grade: 1.5 — Darmstadt, Germany*

- Focus: Object-Oriented Software Engineering & Relational Databases

**Heinrich-Emanuel-Merck-Schule (HEMS)** Sep 2023 – Jun 2025

*State-Certified Technical Assistant for IT | Overall Grade: 1.7 (Year 1: 1.0)*

- Led a 6-week security capstone; engineered an isolated Linux VM network demonstrating security concepts live (Grade: 1.0).
- Authored comprehensive technical documentation and reproducible runbooks.

## 02 // TECHNICAL SKILLS & STACK

**Languages:** C, Python, Java, C++, SQL (SQLite/PostgreSQL), JavaScript, HTML5/CSS3
**Systems & Tools:** Linux/Unix CLI, Bash, Git & GitHub, GDB, Valgrind, VS Code, Vercel, Docker
**Networks & Architecture:** TCP/IP, Subnetting, Routing (Cisco CCNA Core), Memory Management (Pointers, Heap/Stack)
**Languages:** German (Fluent), English (Fluent), Arabic (Native)

## 03 // ENGINEERING PROJECTS

**Personal Interactive Web Platform (ms1a.tech)** JavaScript, HTML5, CSS3, Python/Flask, Git

- Architected a live developer telemetry platform recording focus metrics, study routines, and automated tracking.
- Built responsive, modern UI with vanilla JavaScript and deployed CI/CD pipeline via GitHub Pages.

**CS50x Algorithmic Systems Suite (Harvard University)** C, Memory Architecture, Linux CLI

- Engineered an algorithmic search engine in C, querying 140,000+ words in 0.038s using a Hash Table with collision chaining.
- Eliminated 100% of memory leaks across 1,500+ lines of low-level C code, verifying heap integrity via Valgrind.

**Real-Time Market Arbitrage & Alert Engine** Python, REST APIs, Telegram Bot API

- Engineered an automated data pipeline and scraper monitoring listings to detect arbitrage opportunities in real time.
- Integrated event-driven Telegram alerts delivering instant notifications with pricing metadata and direct links.

## 04 // CERTIFICATIONS

- **Harvard University:** CS50x – Introduction to Computer Science (2026)
- **Cisco Networking Academy:** CCNA – Introduction to Networks (2025)
- **Meta:** Programming Fundamentals in Swift (2025)